02. Dataset Understanding

Understand the original challenge data and the smaller evaluation sample used in the notebook.

Project: Automated Mobile-Technology Brand Sentiment and Headline Generator

Source archive: NLP-Sentiment-Classification-of-Mobile-brands-main.zip | Notebook: main.ipynb | Supporting report and presentation included

Challenge dataset design

The challenge specification describes approximately 4,000 articles and 4,000 tweets in English, Hindi and Hinglish. The original project report says the data was split stratified into 3,000 train, 600 validation and 400 test samples separately for articles and tweets.

Evaluation notebook sample

Subset	Rows in notebook	Mobile-tech predicted
Articles	222	71
Tweets	210	62
Total	432	133

Observed fields

- Text_ID identifies each record with prefixes such as article_ and tweet_.
- Text contains the article or tweet content.
- After preprocessing, the notebook adds detected brands, number of brands, language, Mobile_Tech flag and generated headline/sentiment outputs.
- The final output schema includes Text_ID, Mobile_Tech_Flag, Brands_Entity_Identified and Sentiment_Identified.

Data characteristics

- Tweets are short, noisy and contain mentions, hashtags, URLs, retweets, emojis and informal spelling.
- Articles are longer, can contain multiple paragraphs and may mention several brands in separate contexts.
- Hindi and Hinglish require language detection and translation/transliteration handling.
- Brand sentiment is target-specific: one text may praise one brand and criticize another.

Important availability limitation

The supplied ZIP contains the notebook, report, presentation and task notes, but not the evaluation_data.xlsx file, the original 8,000 labelled training records, source Python modules, or pretrained model weights. The notebook downloads several artifacts from external Google Drive links and imports modules from an AHSG-InterIIT project directory. The complete pipeline cannot be reproduced from this ZIP alone.

Data questions

- Are brand mentions aliases, product names or parent-company names?
- How were sentiment labels defined for sarcasm, comparisons and mixed opinions?

- Are paragraph-level article labels available, or only article-level labels?
- How are multiple occurrences of the same brand aggregated?
- Are translated texts and headlines evaluated against human translations or original-language references?